

Development of Degraders of Cyclin-Dependent Kinases 4 and 6 Based on Rational Drug Design

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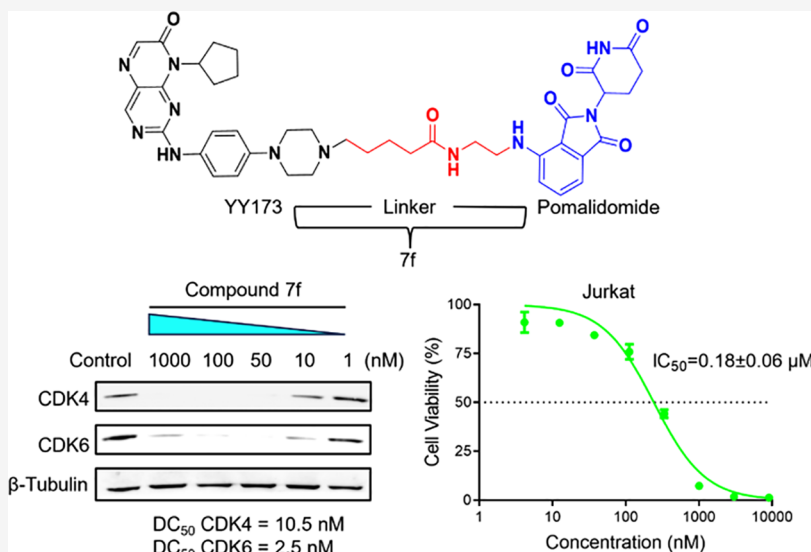
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ABSTRACT: Degradation of target proteins has been considered to be a promising therapeutic approach, but the rational design of compounds for degradation remains a challenge. In this study, we reasonably designed and synthesized only 10 compounds to discover effective CDK4/6 protein degraders. Among the newly synthesized compounds, **7f** achieved dual degradation of CDK4/6 protein, with DC₅₀ values of 10.5 and 2.5 nM, respectively. Compound **7f** also exhibited inhibitory proliferative activity against Jurkat cells with an IC₅₀ value of 0.18 μM. Furthermore, **7f** induced cell apoptosis and G1 phase cell cycle arrest in a dose-dependent manner in Jurkat cells. In conclusion, these findings demonstrate the potential of **7f** as a CDK4/6 degrader and a potential therapeutic strategy against cancer, thereby expanding the potential of CDK4/6 dual PROTACs.

INTRODUCTION

Cyclin-dependent kinases (CDKs) constitute a large family of serine–threonine kinases crucial for regulating cell cycle progression.^{1–3} This family comprises 21 subtypes, categorized into those regulating the cell cycle (CDK1–6 and 14–18) and transcription (CDK7–13, 19, and 20). Despite promising preclinical research results, both first-generation and second-generation CDK inhibitors were discontinued in clinical trials due to nonselective pan-CDK inhibition, resulting in severe cytotoxic effects on normal cells. Among the CDK kinase family, CDK4/6 has been most extensively studied and with great success.^{4,5} CDK4/6 inhibitors have shown promising clinical benefits in recurrent metastatic breast cancer,^{6–8} with four drugs already marketed: palbociclib, ribociclib, abemaciclib, and trilaciclib.^{9–11} Additionally, numerous highly active and selective small-molecule structures have been reported in the

literature and patents.¹² However, acquired drug resistance to CDK4/6 inhibitors has emerged after administration to patients.^{13–15} The molecular mechanism of palbociclib resistance remains unclear.¹⁶ Therefore, the development of novel therapeutic strategies may provide a new solution to the drug resistance.¹⁷

Protac (proteolysis-targeting chimeras) is a heterobifunctional molecule, with one end connected to a ligand that binds the target protein and the other end to an E3 ligase ligand, linked

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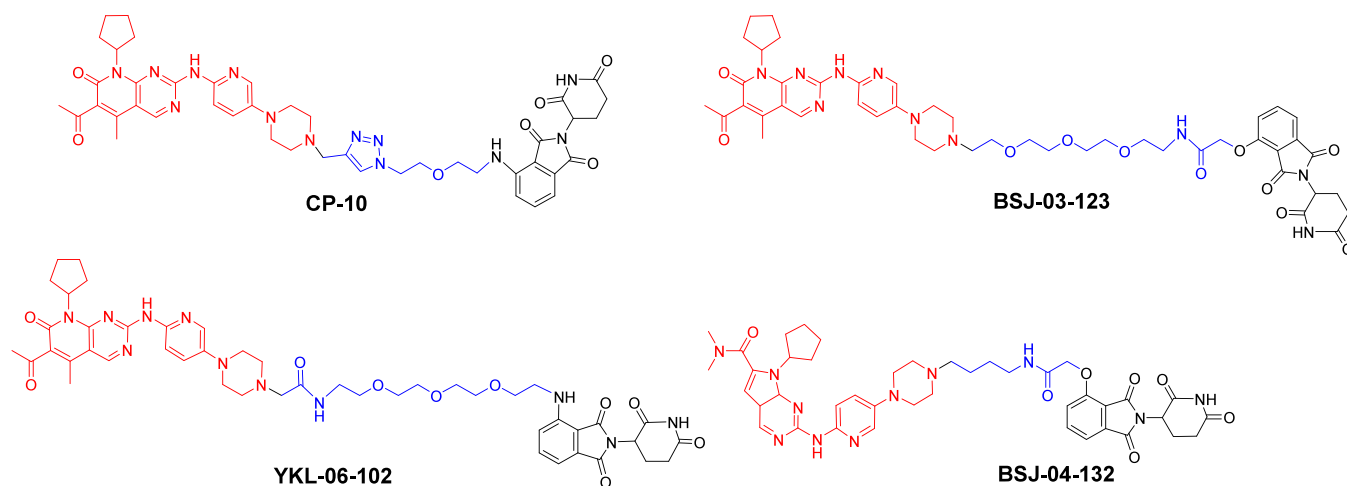


Figure 1. Structures of published CDK4/6-selective PROTACs CP-10, BSJ-03-123, YKL-06-102, and BSJ-04-132.

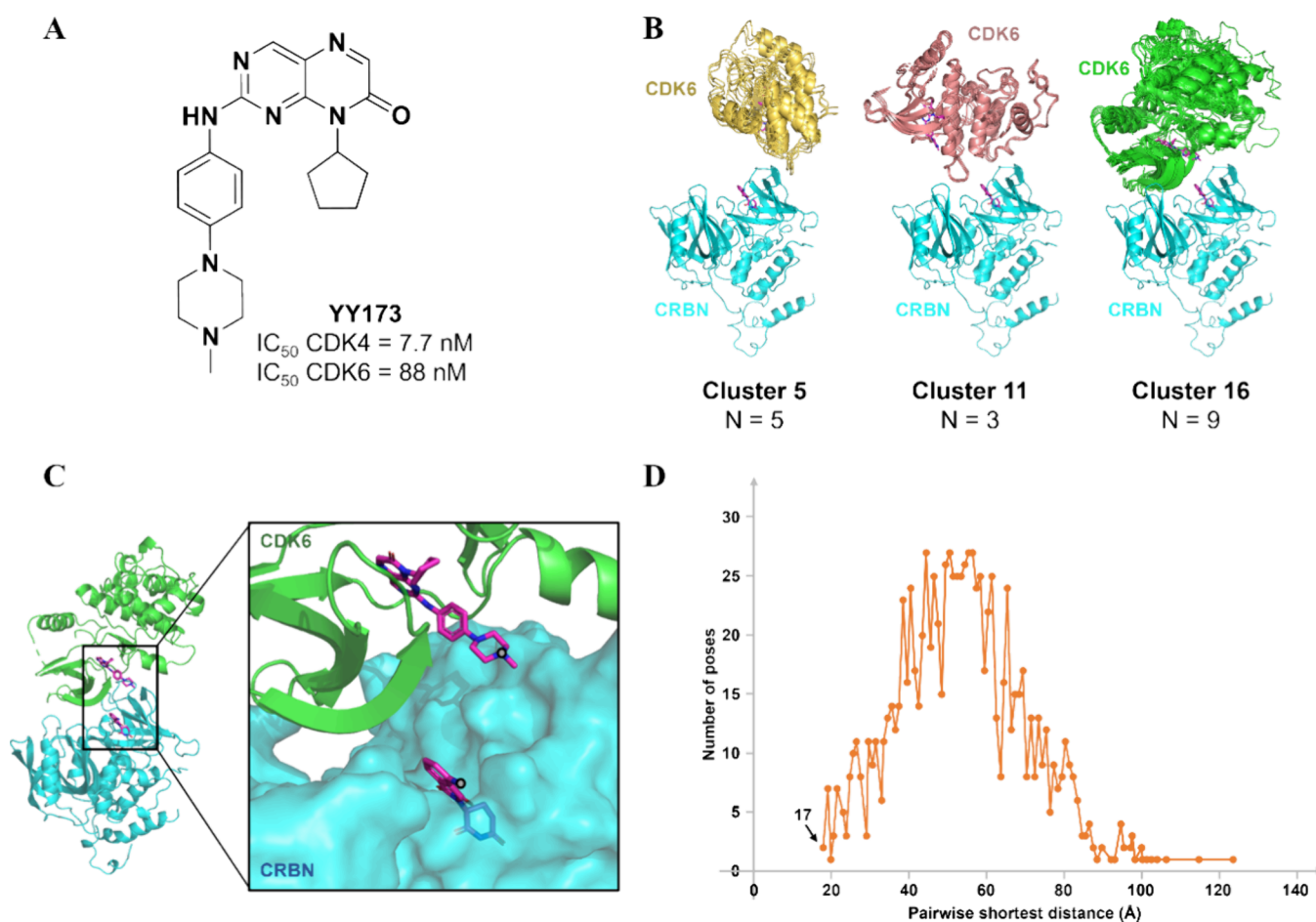
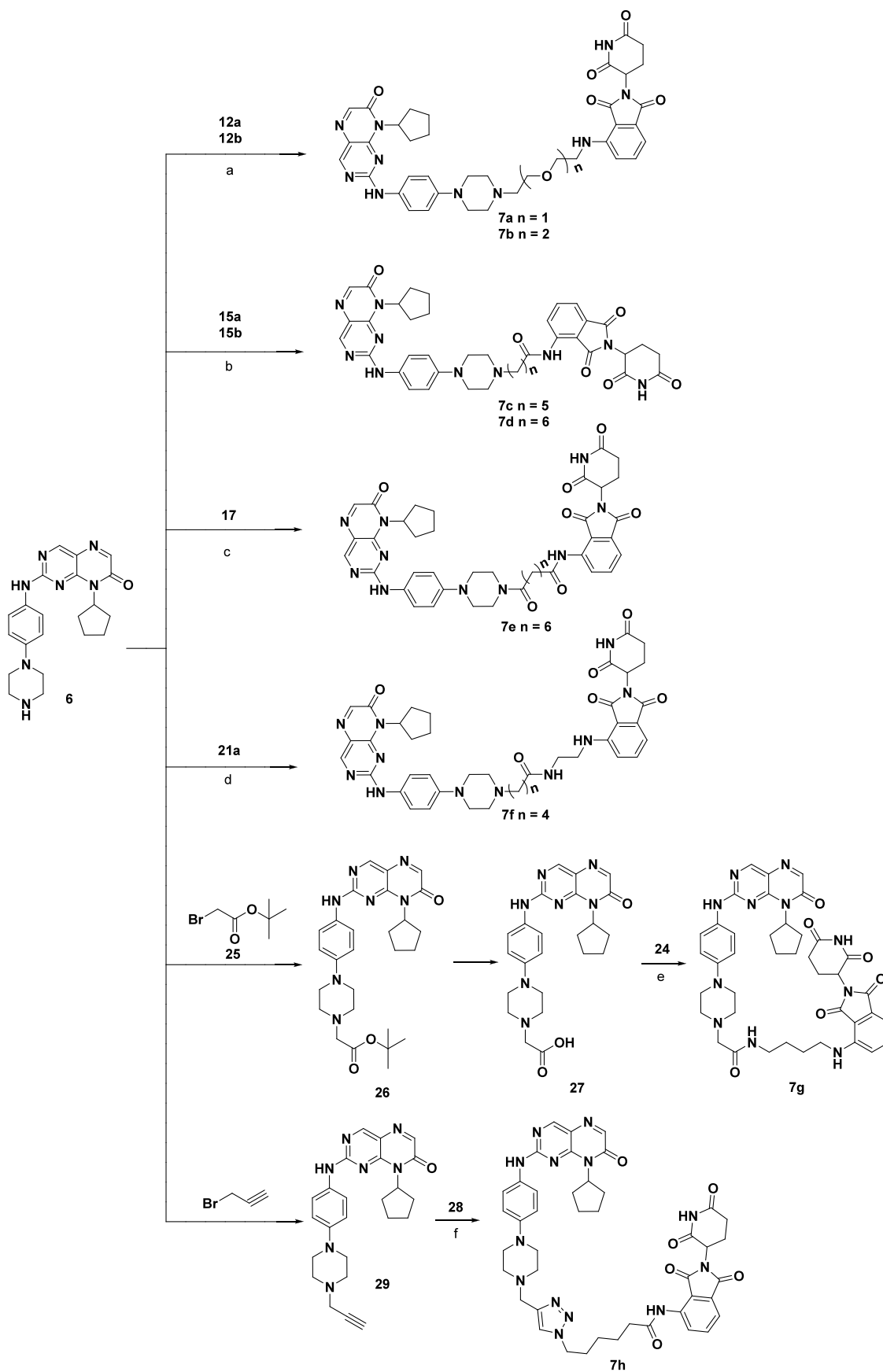


Figure 2. (A) Chemical structure of YY173. (B) Cartoon representations for representative clusters obtained by *k*-means clustering of docking poses between CRBN (PDB: 4CI3) and CDK6 (PDB: 5L2I). (C) Protein docking to design the degrader molecules. Close-up view of the proximity of the YY173 piperazine and pomalidomide. Atoms used for calculation of the pairwise shortest distances between YY173 and pomalidomide are highlighted in black circles. (D) Scatter diagram of the pairwise shortest distances for the 1000 docking poses.

by a suitable linker.^{18,19} The PROTAC molecule binds to both the target protein and E3 ligase, forming a ternary complex. This complex tags the target protein with ubiquitination, leading to its degradation by the intracellular proteasome.^{20–22} PROTACs have the potential to enhance the protein selectivity and overcome resistance induced by target protein mutations or overexpression.^{23,24}

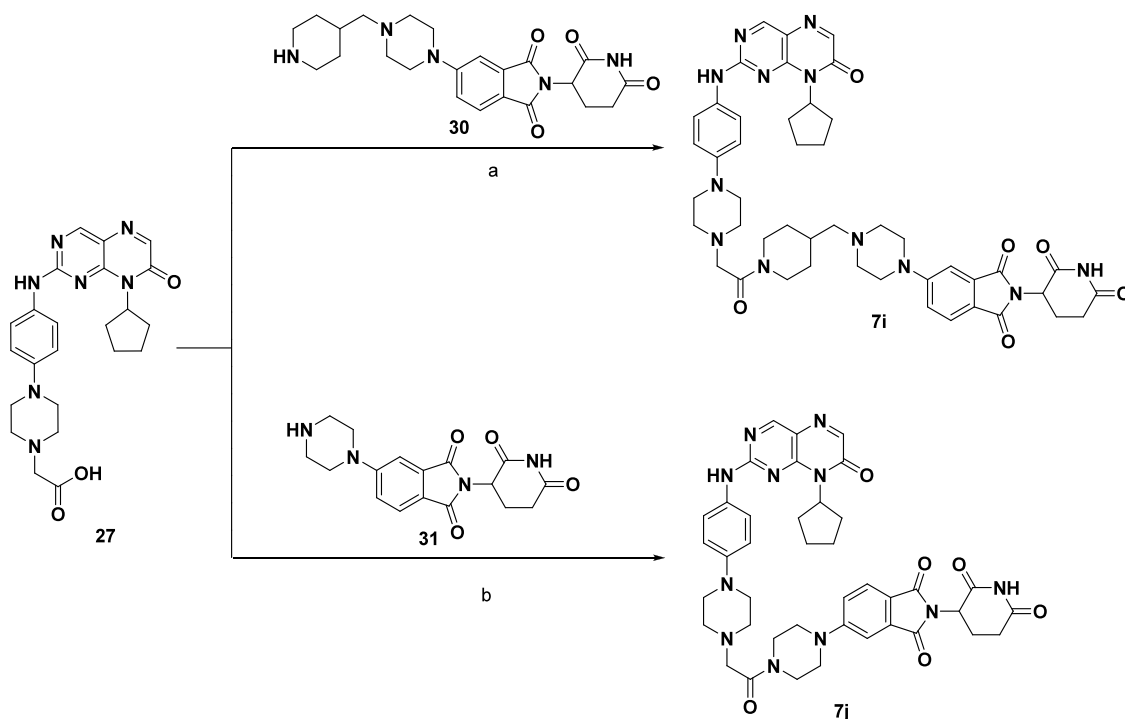
To address the selectivity of CDKs and overcoming drug resistance, great efforts were made to develop CDK4/6 PROTACs. Several CDK6-selective and dual CDK4/6 degraders have been reported,^{25–27} as illustrated in Figure 1. Till now, none of CDK4/6 PROTACs have entered in clinical trials. This may be attributed to the following issues. On the one hand, the majority of current CDK4/6 degraders are derived

Scheme 1. Synthesis Route of Compounds 7a–7h^a

Scheme 1. continued

^aReagents and conditions: (a) DMF DIPEA, 90 °C, 4 h, 9.7–13.4%; (b) MeCN, K₂CO₃, KI, 80 °C, 3 h, 10.5–13.1%; (c) DCM, HOBT, EDCI, RT, 12 h, 17.6%; (d) MeCN, K₂CO₃, KI, 80 °C, 3 h, 11.2%; (e) DMF, DIPEA, HATU, RT, 5 h, 22.2%; (f) *tert*-butanol/DCM(2/1, v/v), sodium ascorbate, copper sulfate aqueous solution, RT, 10 min, 13.6%.

Scheme 2. Synthesis Route of Compounds 7i–7j



^aReagents and conditions: (a) NMP, DIPEA, 150 °C, 150 PSI, 50 w, 30 min, 8.7%; (b) NMP, DIPEA, 150 °C, 150 PSI, 50 W, 30 min, 7.2%.

from palbociclib and ribociclib. Therefore, there is a lack of diversity in small-molecule inhibitors used for developing CDK4/6 degradation agents. On the other hand, the activity validation and optimization of PROTACs incurs high costs in the process of pharmaceutical synthesis and screening.^{28,29} To enhance research efficiency and address these challenges, we propose a novel approach: designing a selective CDK4/6 degradation agent using computational simulation methods.³⁰

In this study, we analyzed the structural characteristics and existing experimental data of CDK4/6 degraders' E3 ligands reported in the current literature. Pomalidomide was selected as the most suitable E3 ligand based on our analysis. Concurrently, we identified compound YY173 (CDK4 IC₅₀ = 7.7 nM, CDK6 IC₅₀ = 88 nM) as a small-molecule inhibitor ligand through screening our in-house database (Figure 2A). YY173 has never been reported as a CDK4/6 protein ligand in the development of CDK PROTACs and would probably exhibit quite different efficiency and selectivity. Using the Rosetta docking method, we simulated the distance between CDK6 and E3 ligase and calculated the optimal length of the linker between YY173 and pomalidomide. Based on this calculation, we designed and synthesized several Protac molecules with different types of linkers. In-depth study proved that most of these compounds could efficiently degrade CDK4/6 at low concentrations. Moreover, our PROTACs exhibited strong antiproliferative effects in Jurkat cells. The rational design and synthesis strategy facilitated the efficient discovery of potent Protac molecules.

RESULTS AND DISCUSSION

Protein Docking Guides PROTAC Design. Previous studies have shown that the binding orientation between the target protein and E3 ligase can vary depending on the number of PROTACs used. Therefore, it is highly desirable to assess potential binding conformations between target protein and E3 ligase to narrow down the chemical search space.³¹ Protein–protein docking-based computational simulations offer an attractive method for this purpose. Consequently, we utilized the Rosetta protein docking framework to simulate possible binding modes and calculate the distance between YY173 and pomalidomide to inform the Protac design.

Initially, using the complex structure of pomalidomide-bound CRBN and YY173-bound CDK6, we conducted a global docking experiment employing Rosetta docking, generating 1000 models. The interaction between CRBN and CDK6 yielded multiple low-energy minima. Considering that the docking experiment sampled a range of low-energy binding conformations, we clustered the top 200 conformations and then provided a set of feasible binding modes. Figure 2B shows three representative clusters. While the accuracy of docking in predicting binding modes remains to be demonstrated, the overall conformational landscape provides a rationale for determining the required minimum linker lengths and suggests suitable linkage positions (Figure 2).

To evaluate whether the docking information could guide the degrader design, we calculated the pairwise shortest distances between the selected solvent-exposed atoms of YY173 and

Table 1. Structures and DC₅₀ Values of Compounds 7a–7j from YY173 and Pomalidomide

Compound	Linker	DC ₅₀ (nM) CDK4	DC ₅₀ (nM) CDK6	Dmax(%) CDK4	Dmax(%) CDK6
7a		161.9	9.5	60	91
7b		557.2	5.3	53	94
7c		14.9	58.0	95	85
7d		>1000	6.6	41	95
7e		434.1	43.3	60	92
7f		10.5	2.5	100	96
7g		>1000	>1000	14	40
7h		>1000	16.3	42	95
7i		137.8	2.3	68	90
7j		9.8	3.9	75	86

pomalidomide for all poses (Figure 2C). This analysis revealed distances ranging from 17 to 18 Å (Figure 2D), corresponding to a linker length of six to eight atoms, which was sufficient to bridge the two moieties. In theory, the shortest possible linker for a ligase–target pair should provide the most selective compound, as it would restrict the degrees of freedom for binding of CDK6 to CRBN.

Based on our simulations, we concentrated on the six to eight carbon atoms length linkers, significantly reducing our synthetic workload. After analyzing the published types of linkers, we designed and synthesized a series of compounds incorporating various types of linkers, including amides, alkyl chains, ethers, and piperazine chains.

Chemistry. The synthesis of compounds **7a–7h** is outlined in [Scheme 1](#), while the synthesis of key intermediates **6**, **27**, and **29** is presented in [Scheme S1](#). Additionally, the synthesis of the Protac linker with the E3 ligand is detailed in [Scheme S2](#). Intermediate **6** undergoes a DIPEA-mediated nucleophilic substitution reaction with **12a** and **12b** to yield the required compounds **7a** and **7b**. Under alkaline catalysis using K_2CO_3 and KI, intermediate **6** undergoes a nucleophilic substitution reaction with **15a**, **15b**, and **21a** to produce compounds **7c**, **7d**, and **7f**, respectively. The key intermediate **6** reacts with **17** under the condensation initiation of HOBT and EDCI to yield **7e**. Furthermore, the amide coupling reaction, mediated by HATU and DIPEA, yielded the required compound **7g**. For **7h**, we

selected intermediates **29** and **28** with azide heads to undergo a click reaction, yielding the desired compound.

The synthesis of **7i** and **7j** is illustrated in [Scheme 2](#). The key intermediates **27** and **30** or **31** were reacted via microwave-assisted reactions at 150 PSI and 50 W in an alkaline environment provided by DIPEA to yield compounds **7i** and **7j**.

Degradation Activity Study. We assessed the degradation potential of all designed CDK4/6-targeting PROTACs through Western blot assay in Jurkat cells, which express both CDK4 and CDK6 proteins,²⁹ at concentrations of 1, 10, 50, 100, and 1000 nM (Figure S1). As depicted in Table 1, almost all compounds demonstrated superior degradation of CDK6 protein than CDK4. Compounds 7b and 7d exhibited increased activity against CDK6 compared to 7a and 7c, suggesting that a slight extension of the linker is beneficial for degradation activity. Therefore, we focused on linker with eight to nine carbon atoms in length. Compounds 7e, 7f, and 7g were connected by an amide bond. Among them, 7f demonstrated the most significant degradation effect against CDK4/6 protein, with DC₅₀ values of 10.5 and 2.5 nM, respectively. Compound 7f also exhibited maximum degradation efficiencies of 100 and 96% against CDK4 and CDK6, respectively. These data indicate that our prediction of the linker length is accurate, reducing the need to explore lengths of different linker classes extensively. Subsequently, we synthesized several additional linkers commonly found in the literature. Compound 7i showed the highest

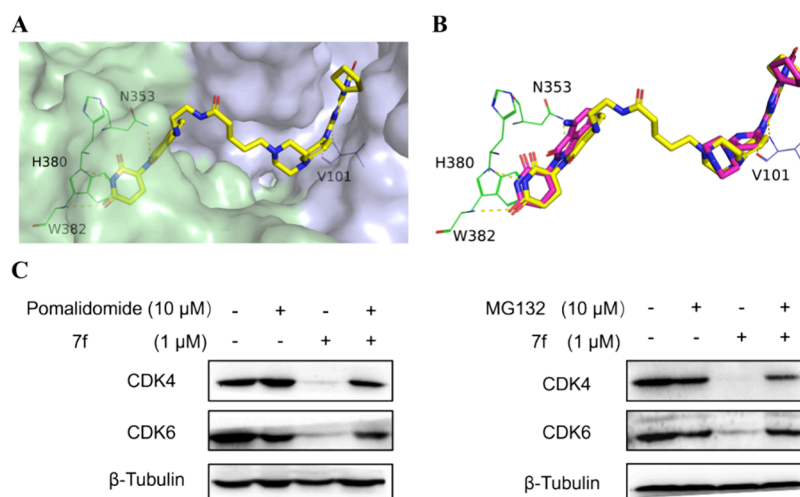


Figure 3. (A) Predicted binding mode of compound **7f** in the interface between CDK6 (PDB code 5L2I) and the CRBN kinase domain (PDB code 4CI3). (B) Overlaid model of compound **7f** (yellow) and raw ligands (magentas). (C) Mechanistic study of CDK4/6 degradation induced by **7f** in Jurkat cells. Cells were pretreated with MG132 and pomalidomide to insulated them from CDK4/6 degradation by **7f**.

degradation efficiency against the CDK6 protein, with a DC_{50} value of 2.3 nM, and also showed selectivity for the CDK4 protein. Compound **7i** played a significant guiding role in the research of the selective CDK6 degraders. We selected compound **7f** for further study as it exhibits great degradation activity against both CDK4 and CDK6, which was in accordance with CDK-targeting drugs in clinical use.³²

Binding Mode Analysis of Compound 7f. To explore the molecular mechanism of degraders, we employed computational simulation to predict the binding mode of compound **7f**, aiming to elucidate the structure–activity relationship of YY173 and verify the feasibility of reasonable design based on computational simulation. Leveraging these protein–protein docking results, which revealed the shortest linker length configuration, we proceeded to predict the binding mode of **7f**. Utilizing the conformation generation software Cyndi,³³ we generated 1000 conformations of small molecules, obtaining diverse results. Subsequently, we superimposed these conformations onto YY173 and pomalidomide using the SHAFT program,³⁴ selecting the optimal conformation of **7f**. Finally, we optimized the structure of the ternary complex system using Maestro. The results revealed that **7f** binds to the groove at the interface between CDK6 and the CRBN kinase domain (Figure 3A), maintaining the binding conformation of ligands (YY173 and pomalidomide) and polar interactions with protein residues (N353, W382, H380 in CRBN, and V101 in CDK6) (Figure 3B). Additionally, we validated that the degradation of CDK4/6 by **7f** is ligand-dependent and proteasome-dependent (Figure 3C). We selected pomalidomide as a ligand for E3 ubiquitin ligase in the development of our target compounds. Therefore, we chose pomalidomide to compete with **7f** for binding with CRBN. The result showed that excess pomalidomide blocks the degradation of CDK4/6. Besides, the proteasome inhibitor MG132 also blocks the degradation of CDK4/6 induced by **7f**, indicating that the degradation is proteasome-dependent. These findings indicated that compound **7f** could effectively induce the degradation of the CDK4/6 protein through the ubiquitin-proteasome pathway.

Cell Growth Inhibition Activity of 7f in Jurkat Cells. We further evaluated the proliferative effects of **7f** against Jurkat cells using palbociclib and YY173 as the positive control. As shown in Table 2, the IC_{50} values of palbociclib and YY173 in Jurkat cell

were 1.495 and 1.466 μ M, respectively. However, compound **7f**, with better degradation activity, demonstrated an eightfold improvement in antiproliferative effect compared with YY173 in Jurkat cells with IC_{50} values of 0.183 μ M.

Compound 7f Is a Selective CDK4/6 Degradator. We then investigated the degradation mechanism of compound **7f** on CDK4/6 protein through a Western blot assay. First, **7f** exhibited concentration-dependent degradation of CDK4/6 protein (Figure 4A). Additionally, **7f** showed the time-dependent degradation of CDK4/6 proteins. In Jurkat cells, protein degradation of CDK4/6 commenced after 3 h with 1 μ M and complete degradation of CDK4/6 was observed after 12 h (Figure 4B). Furthermore, a selective degradation evaluation was conducted, revealing that **7f** exhibited a high selectivity for the degradation of CDK4/6. At a concentration of 1 μ M, **7f** induced complete degradation of CDK4/6 proteins while having little or no effect on the protein levels of CDK1, CDK2, CDK4, CDK5, CDK7, and CDK8 (Figure 4C). In conclusion, **7f** demonstrates efficient and selective degradation characteristics against CDK4/6.

Table 2. Evaluation of the Antiproliferative Activities of Compound **7f** against Jurkat Cells

compound	in vitro cytotoxicity IC_{50} (μ M)
	Jurkat
7f	0.183 $\pm$ 0.061
YY173	1.466 $\pm$ 0.348
palbociclib	1.495 $\pm$ 0.246

Evaluation of the Antitumor Effects of 7f in Jurkat Cells. Given that the CDK4/6 degrader **7f** exhibits effective antiproliferative activity *in vitro*, flow cytometry was selected to further evaluate the effects of **7f** on cell cycle arrest and apoptosis induction in Jurkat cells. As depicted in Figure 5A, compound **7f** induced apoptosis and arrested cell division in the G1 phase of Jurkat cells in a concentration-dependent manner.

To elucidate the potential mechanism of **7f**-induced cell apoptosis, apoptosis-related proteins were examined via a Western blot assay. The results revealed that the cleavage of PARP and Caspase-3 in Jurkat cells increased in a dose-

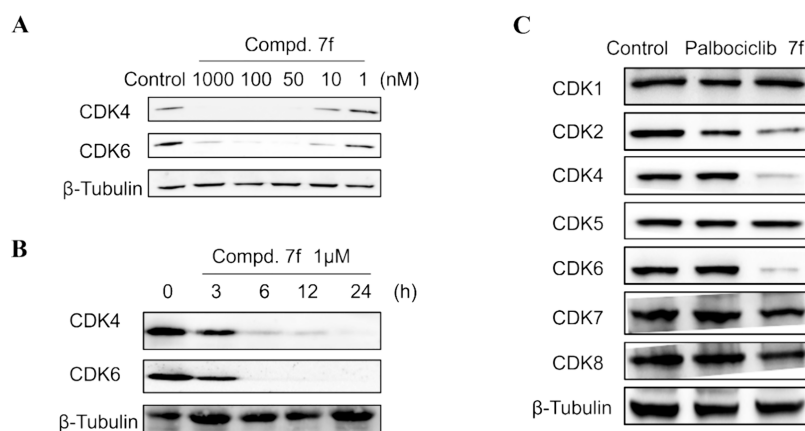


Figure 4. (A, B) 7f-induced concentration and time-dependent degradation of CDK4/6 protein in Jurkat cells. (C) Western blot analysis of 7f selectivity.

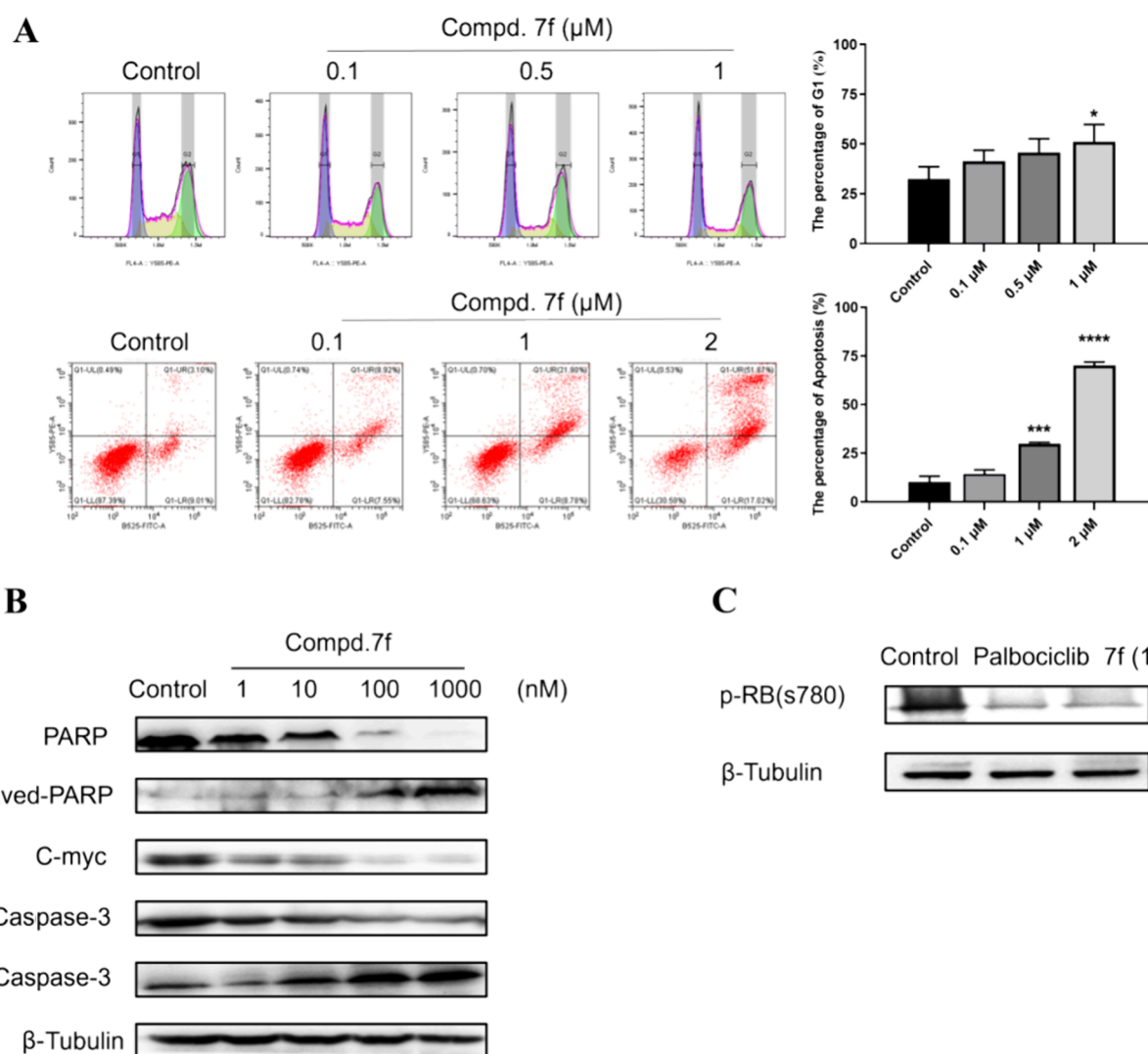


Figure 5. (A) Cell cycle and apoptosis analysis of 7f in Jurkat cells. (B) The effects on cleavage of PARP and Caspase-3 were detected by Western blot. (C) Compound 7f inhibits the phosphorylation of RB. (* P < 0.05, *** P < 0.001, **** P < 0.0001, compared with control).

dependent manner, indicating that 7f induces cell apoptosis by up-regulation of cleaved Caspase-3 and cleaved PARP proteins (Figure 5B). CDK4/6 mainly regulates the G1-S phases through the CDK4/6-RB pathway, promoting cell proliferation. Therefore, we evaluated the ability of 7f to block the phosphorylation

of RB (Ser780) in Jurkat cells. Our findings demonstrated that 7f effectively suppresses the expression of p-RB (Ser780) (Figure 5C).

CONCLUSIONS

Given the potential for multiple binding conformations between the target protein and E3 ubiquitin ligase, we employed a rational drug design based on computational simulations of protein docking. Drawing from our methodology and relevant literature, we designed and synthesized 10 target compounds. We then evaluated the degradation activity and DC₅₀ values of these compounds against CDK4/6. Among the synthesized compounds, **7f** emerged as the most promising candidate, demonstrating high activity, selectivity, and rapid induction of the CDK4/6 protein degradation. Mechanistic studies indicated that the degradation effect of **7f** on CDK4/6 was achieved through the ubiquitin–proteasome pathway. *In vitro* activity evaluation further showed that **7f** effectively inhibited cell proliferation, leading to cell cycle G1 arrest and the induction of apoptosis. Furthermore, we examined the pharmacokinetic properties of compound **7f** and found that it lacks the potential to be developed as an oral drug (Table S1). Although our work provided innovative strategies in linker design for PROTACs, it still needs more efforts on improving the drug-like properties of PROTACs.

EXPERIMENTAL SECTION

Chemistry. All chemical reagents and solvents used in the experiments were obtained from commercial sources (Bidepharm, LinkChem, and WuXi AppTec) and were not subjected to further purification. All solvents and chemicals used were of reagent grade. Silica gel (200–300 mesh) used for column chromatography and thin-layer chromatography (TLC) used to monitor progress of the reaction were purchased from Qingdao Haiyang Chemical Company. ¹H NMR spectra were performed on a Bruker 400 MHz/AVANCE III 400. The low-resolution mass spectra (LC-MS) were measured with Agilent 6120 LC-MS. The high-resolution mass spectra (HRMS, Waters XEVO G2 TOF) were measured at the Analysis and Test Center of ECUST. The purity of final compounds was analyzed by HPLC (Waters ACQUITY Arc). The HPLC instrument was equipped with a 2998 PDA detector and a Zorbax RP-18 column (2.7 μm, 4.6 mm × 50 mm, reverse-phase column). The mobile phase consisted of a mobile phase of 95% A (CH₃CN) and a mobile phase of 5% B (0.1% CF₃COOH in water). The flow rate was 0.5 mL/min. All compounds are >95% pure by HPLC analysis.

4-((2-(2-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)-amino)phenyl)piperazin-1-yl)ethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (**7a**). The compounds 2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl 4-methylbenzenesulfonate (**12a**) (70.00 mg, 0.14 mmol) and 8-cyclopentyl-2-((4-(piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (**6**) (50.00 mg, 0.13 mmol) were dissolved in DMF (2 mL), and then *N,N*-diisopropylethylamine (0.3 mL) was added. The reaction was stirred at 90 °C for 4 h and then washed with water, and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 35/1, v/v) to give compound (**7a**) (12.60 mg, 13.4%) as yellow solid. Melting point: 198.7–199.8 °C. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.10 (s, 1H), 9.95 (s, 1H), 8.76 (s, 1H), 7.83 (s, 1H), 7.59 (dd, *J* = 8.6, 7.0 Hz, 1H), 7.49 (d, *J* = 8.6 Hz, 2H), 7.16 (d, *J* = 8.6 Hz, 1H), 7.04 (d, *J* = 7.0 Hz, 1H), 6.89 (d, *J* = 8.7 Hz, 2H), 6.62 (t, *J* = 5.8 Hz, 1H), 5.73–5.52 (m, 1H), 5.05 (dd, *J* = 12.8, 5.3 Hz, 1H), 3.61 (dt, *J* = 11.1, 5.5 Hz, 4H), 3.49 (q, *J* = 5.5 Hz, 2H), 3.04 (t, *J* = 5.1 Hz, 4H), 2.55 (t, 4H), 2.27–2.17 (m, 2H), 2.03–1.95 (m, 2H), 1.92–1.82 (m, 2H), 1.81–1.71 (m, 2H), 1.63–1.52 (m, 2H). ¹³C NMR (151 MHz, DMSO-*d*₆) δ 172.77, 170.06, 168.96, 167.30, 158.61, 156.55, 150.24, 147.36, 146.47, 136.22, 132.08, 130.90, 121.78, 117.54, 110.66, 109.22, 68.81, 68.45, 57.24, 53.12, 48.63 (d, *J* = 18.2 Hz), 41.66, 31.54, 31.29, 30.98, 30.38, 29.43, 28.99, 27.31, 25.01, 22.16, 20.77, 14.09. HRMS (ESI): (*m/z*): [M + H]⁺ calcd for C₃₈H₄₂N₁₀O₆, 734.3289; found 735.3369. HPLC purity: 98.65%, retention time = 1.450 min.

4-((2-(2-(2-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)-amino)phenyl)piperazin-1-yl)ethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (**7b**). General procedure for the synthesis of compound **7b**. Melting point: 265.5–266.9 °C. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.10 (s, 1H), 9.94 (s, 1H), 8.76 (s, 1H), 7.83 (s, 1H), 7.58 (dd, *J* = 8.6, 7.1 Hz, 1H), 7.49 (d, *J* = 8.8 Hz, 2H), 7.15 (d, *J* = 8.6 Hz, 1H), 7.04 (d, *J* = 7.0 Hz, 1H), 6.90 (d, *J* = 8.6 Hz, 2H), 6.62 (t, *J* = 5.8 Hz, 1H), 5.73–5.53 (m, 1H), 5.05 (dd, *J* = 12.9, 5.4 Hz, 1H), 3.64 (t, *J* = 5.4 Hz, 2H), 3.60–3.56 (m, 2H), 3.56–3.51 (m, 4H), 3.48 (q, *J* = 5.4 Hz, 2H), 3.08–2.99 (m, 4H), 2.56–2.53 (m, 4H), 2.26–2.18 (m, 2H), 2.04–1.97 (m, 2H), 1.93–1.81 (m, 2H), 1.80–1.72 (m, 2H), 1.62–1.52 (m, 2H). ¹³C NMR (151 MHz, DMSO-*d*₆) δ: 172.96, 170.17, 168.98, 167.33, 159.37, 158.63, 156.58, 146.46, 136.27, 132.11, 121.82, 117.50, 115.57, 110.71, 109.26, 69.78, 69.75, 68.92, 68.42, 57.26, 53.11, 48.72, 48.61, 41.79, 31.03, 27.33, 22.21. HRMS (ESI): (*m/z*): [M + H]⁺ calcd for C₄₀H₄₆N₁₀O₇, 778.3551; found 779.3631. HPLC purity: 98.94%, retention time = 1.490 min.

6-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)hexanamide (**7c**). To the solution of 6-bromo-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)hexanamide (**15a**) (91.00 mg, 0.20 mmol) and 8-cyclopentyl-2-((4-(piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (**6**) (50.00 mg, 0.13 mmol) in acetonitrile (1 mL) were added K₂CO₃ (89.00 mg, 0.64 mmol) and KI (20.00 mg, 0.12 mmol). The reaction was stirred at 80 °C for 3 h and then washed with water, and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 40/1, v/v) gave compound **7c** (12.70 mg, 13.1%) as a yellow solid. Melting point: 202.3–204.5 °C. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.16 (s, 1H), 9.94 (s, 1H), 9.71 (s, 1H), 8.76 (s, 1H), 8.48 (d, *J* = 8.4 Hz, 1H), 7.95–7.75 (m, 2H), 7.61 (d, *J* = 7.2 Hz, 1H), 7.50 (d, *J* = 8.8 Hz, 2H), 6.92 (d, *J* = 9.0 Hz, 2H), 5.65–5.60 (m, 1H), 5.15 (dd, *J* = 12.7, 5.4 Hz, 1H), 3.37 (s, 4H), 3.09 (s, 4H), 2.61–2.58 (m, 2H), 2.35–2.30 (m, 2H), 2.23–2.20 (m, 4H), 2.01–1.98 (m, 2H), 1.94–1.72 (m, 4H), 1.70–1.64 (m, 2H), 1.61–1.57 (m, 2H), 1.53–1.45 (m, 2H), 1.38–1.30 (m, 2H). ¹³C NMR (151 MHz, DMSO-*d*₆) δ: 174.67, 173.17, 172.42, 170.19, 168.11, 167.07, 159.68, 159.00, 156.94, 136.95, 136.52, 131.87, 130.05, 128.99, 126.69, 122.15, 118.70, 117.37, 115.97, 49.33, 31.68, 31.33, 29.36, 29.22, 29.08, 27.69, 22.38. HRMS (ESI): (*m/z*): [M + H]⁺ calcd for C₄₀H₄₄N₁₀O₆, 760.3445; found 761.3526. HPLC purity: 96.83%, retention time = 1.560 min.

7-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)heptanamide (**7d**). General procedure for the synthesis of compound **7d**. Melting point: 212.6–213.8 °C. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.16 (s, 1H), 9.94 (s, 1H), 9.69 (s, 1H), 8.75 (s, 1H), 8.47 (d, *J* = 8.4 Hz, 1H), 7.88–7.78 (m, 2H), 7.60 (d, *J* = 7.3 Hz, 1H), 7.49 (d, *J* = 8.5 Hz, 2H), 6.91 (d, *J* = 8.8 Hz, 2H), 5.72–5.50 (m, 1H), 5.15 (dd, *J* = 12.8, 5.4 Hz, 1H), 3.34 (t, 4H), 3.07 (t, *J* = 4.7 Hz, 2H), 2.67–2.52 (m, 2H), 2.46–2.43 (m, 2H), 2.30 (t, *J* = 7.3 Hz, 2H), 2.46–2.17 (m, 2H), 2.11–1.94 (m, 2H), 1.94–1.81 (m, 2H), 1.80–1.71 (m, 2H), 1.68–1.61 (m, 2H), 1.60–1.53 (m, 2H), 1.50–1.42 (m, 2H), 1.38–1.30 (m, 4H). ¹³C NMR (151 MHz, DMSO-*d*₆) δ: 172.76, 172.04, 169.79, 167.69, 166.67, 159.34, 158.60, 156.55, 147.38, 136.55, 136.10, 131.46, 126.30, 121.75, 118.30, 116.99, 115.56, 57.86, 52.79, 48.91, 48.73, 36.50, 30.92, 28.45, 27.29, 26.69, 26.14, 24.78, 21.97. HRMS (ESI): (*m/z*): [M + H]⁺ calcd for C₄₁H₄₆N₁₀O₆, 774.3602; found 775.3685. HPLC purity: 96.67%, retention time = 1.500 min.

8-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)-8-oxooctanamide (**7e**). To the solution of 8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-8-oxooctanoic acid (**17**) (50.00 mg, 0.12 mmol) and 8-cyclopentyl-2-((4-(piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (**6**) (30.00 mg, 0.08 mmol) in dichloromethane (1 mL) were added EDCI (20.00 mg, 0.10 mmol) and HOBT (15.00 mg, 0.11 mmol) at room temperature. The reaction was stirred overnight and then washed with water, and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 100/1, v/v) gave compound (**7e**) (10.8 mg, 17.6%) as a yellow solid. Melting point: 221.4–222.3 °C. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.15 (s, 1H), 9.97 (s, 1H), 9.70 (s, 1H), 8.77 (s, 1H),

8.47 (d, $J = 8.4$ Hz, 1H), 7.84 (d, $J = 1.8$ Hz, 1H), 7.81 (d, $J = 8.1$ Hz, 1H), 7.60 (d, $J = 7.3$ Hz, 1H), 7.53 (d, $J = 8.7$ Hz, 2H), 6.96 (d, $J = 9.0$ Hz, 2H), 5.73–5.47 (m, 1H), 5.15 (dd, $J = 12.8, 5.4$ Hz, 1H), 3.59 (t, $J = 5.4$ Hz, 4H), 3.14–2.99 (m, 4H), 2.68–2.52 (m, 2H), 2.48–2.44 (m, 2H), 2.35 (t, $J = 7.4$ Hz, 2H), 2.28–2.18 (m, 2H), 2.12–1.99 (m, 2H), 1.89 (d, $J = 12.8$ Hz, 2H), 1.82–1.73 (m, 2H), 1.67–1.57 (m, 4H), 1.54–1.48 (m, 2H), 1.36–1.32 (m, 4H). ^{13}C NMR (151 MHz, DMSO- d_6) δ 172.76, 172.03, 170.62, 169.79, 167.69, 166.67, 159.34, 158.56, 156.53, 136.55, 136.10, 131.46, 126.28, 121.67, 116.98, 116.27, 49.42, 48.99, 48.91, 44.80, 36.48, 32.15, 30.92, 28.46, 28.33, 27.29, 24.68, 24.63, 21.97. HRMS (ESI): (m/z): [$M + H$] $^+$ calcd for $\text{C}_{42}\text{H}_{46}\text{N}_{10}\text{O}_7$, 802.3551; found 803.3625. HPLC purity: 98.89%.

5-(4-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)-N-(2-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethylpentanamide (**7f**). Synthesis steps refer to the synthesis of **7c**. Melting point: 185.2–186.6 °C. ^1H NMR (400 MHz, DMSO- d_6) δ 11.10 (s, 1H), 9.94 (s, 1H), 8.76 (s, 1H), 8.04 (t, $J = 5.7$ Hz, 1H), 7.83 (s, 1H), 7.59 (dd, $J = 8.6, 7.1$ Hz, 1H), 7.50 (d, $J = 8.6$ Hz, 2H), 7.18 (d, $J = 8.7$ Hz, 1H), 7.03 (d, $J = 7.0$ Hz, 1H), 6.92 (d, $J = 8.8$ Hz, 2H), 6.73 (t, $J = 6.1$ Hz, 1H), 5.72–5.50 (m, 1H), 5.05 (dd, $J = 12.9, 5.4$ Hz, 1H), 3.42–3.35 (m, 4H), 3.31–3.21 (m, 4H), 3.07 (t, 4H), 2.61–2.53 (m, 2H), 2.35–2.26 (m, 2H), 2.25–2.16 (m, 2H), 2.11–2.06 (m, 2H), 2.05–1.95 (m, 2H), 1.94–1.82 (m, 2H), 1.81–1.71 (m, 2H), 1.58 (t, $J = 5.8$ Hz, 2H), 1.51 (q, $J = 7.4$ Hz, 2H), 1.45–1.39 (m, 2H). ^{13}C NMR (151 MHz, DMSO- d_6) δ : 172.82, 172.71, 170.09, 168.71, 167.31, 159.34, 158.60, 156.55, 147.37, 146.39, 136.20, 132.21, 121.76, 117.18, 115.56, 110.53, 109.23, 57.57, 54.93, 52.73, 48.71, 48.53, 41.56, 37.98, 35.21, 30.98, 29.60, 27.30, 25.83, 23.18, 22.18. HRMS (ESI): (m/z): [$M + H$] $^+$ calcd for $\text{C}_{41}\text{H}_{47}\text{N}_{11}\text{O}_6$, 789.3711; found 790.3786. HPLC purity: 97.35%.

2-(4-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)-N-(4-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butylacetamide (**7g**). Compounds 2-(4-(4-((8-cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)acetic acid (**27**) (52.00 mg, 0.12 mmol) and 4-((4-aminobutyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (**24**) (20.00 mg, 0.06 mmol) were dissolved in DMF (1 mL), and then N,N -diisopropylethylamine (0.3 mL) and HATU (26 mg, 0.07 mmol) were added. The reaction was stirred at room temperature for 5 h and then washed with water, and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 50/1, v/v) to give compound (**7g**) (10.00 mg, 22.2%) as yellow solid. Melting point: 275.7–276.9 °C. ^1H NMR (600 MHz, DMSO- d_6) δ 11.08 (s, 1H), 9.95 (s, 1H), 8.76 (s, 1H), 7.83 (s, 2H), 7.57 (dd, $J = 8.6, 7.0$ Hz, 1H), 7.50 (d, $J = 8.5$ Hz, 2H), 7.11 (d, $J = 8.6$ Hz, 1H), 7.01 (d, $J = 7.0$ Hz, 1H), 6.91 (d, $J = 8.7$ Hz, 2H), 6.57 (t, $J = 6.1$ Hz, 1H), 5.72–5.51 (m, 1H), 5.04 (dd, $J = 12.8, 5.5$ Hz, 1H), 3.32–3.30 (m, 2H), 3.17–3.14 (m, 2H), 3.11 (t, $J = 5.3$ Hz, 4H), 2.96 (s, 2H), 2.56 (t, $J = 4.9$ Hz, 4H), 2.53–2.51 (m, 2H), 2.26–2.19 (m, 2H), 2.04–1.95 (m, 2H), 1.94–1.81 (m, 2H), 1.80–1.73 (m, 2H), 1.60–1.54 (m, 4H), 1.53–1.48 (m, 2H). ^{13}C NMR (151 MHz, DMSO- d_6) δ 172.82, 170.11, 168.93, 167.30, 159.35, 158.61, 156.56, 147.28, 146.40, 136.29, 132.22, 121.78, 117.23, 115.64, 110.37, 109.02, 61.33, 52.87, 48.55, 41.52, 40.00, 37.85, 30.98, 26.71, 26.22, 25.02, 22.16. HRMS (ESI): (m/z): [$M + H$] $^+$ calcd for $\text{C}_{40}\text{H}_{45}\text{N}_{11}\text{O}_6$, 775.3554; found 776.3636. HPLC purity: 99.69%.

6-(4-(4-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)hexanamide (**7h**). Compounds 8-cyclopentyl-2-((4-(4-(prop-2-yn-1-yl)piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (**29**) (30.00 mg, 0.07 mmol), 6-azido-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)-hexanamide (**28**) (34.80 mg, 0.08 mmol), and sodium ascorbate (42.00 mg, 0.21 mmol) were dissolved in a mixture of *tert*-butanol/dichloromethane (2/1, v/v) (4 mL), and then aqueous copper sulfate (1 mL) was added. The reaction was stirred at room temperature for 10 min and then washed with water, and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 50/1, v/v) to give compound (**7h**) (8.00 mg, 13.6%) as orange solid. Melting point: 189.6–190.2 °C. ^1H NMR (400 MHz, DMSO- d_6) δ 11.16 (s, 1H), 9.94 (s, 1H), 9.70 (s, 1H), 8.75 (s, 1H),

8.45 (d, $J = 8.4$ Hz, 1H), 8.03 (s, 1H), 7.82 (s, 1H), 7.80 (d, $J = 7.9$ Hz, 1H), 7.61 (d, $J = 7.3$ Hz, 1H), 7.49 (d, $J = 8.5$ Hz, 2H), 6.91 (d, $J = 8.6$ Hz, 2H), 5.72–5.51 (m, 1H), 5.14 (dd, $J = 12.8, 5.4$ Hz, 1H), 4.35 (t, $J = 7.1$ Hz, 2H), 3.62 (s, 2H), 3.19–2.99 (m, 6H), 2.60–2.53 (m, 4H), 2.48–2.44 (m, 2H), 2.22 (t, $J = 10.0$ Hz, 2H), 2.13–1.95 (m, 2H), 1.87 (q, $J = 7.4$ Hz, 4H), 1.79–1.71 (m, 2H), 1.65 (t, $J = 7.5$ Hz, 2H), 1.58–1.55 (m, 2H), 1.32–1.30 (m, 2H). ^{13}C NMR (151 MHz, DMSO- d_6) δ : 172.77, 171.87, 169.80, 167.65, 166.67, 159.31, 158.60, 156.55, 136.50, 136.09, 131.47, 126.35, 121.76, 118.34, 117.04, 115.68, 57.55, 49.17, 48.91, 36.18, 30.93, 29.40, 27.29, 24.11, 23.05, 21.98, 19.20, 13.50. HRMS (ESI): (m/z): [$M + H$] $^+$ calcd for $\text{C}_{43}\text{H}_{47}\text{N}_{13}\text{O}_6$, 841.3772; found 842.3853. HPLC purity: 99.46%, retention time = 1.380 min.

5-(4-(1-(2-(4-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)acetyl)piperidin-4-yl)methyl)piperazin-1-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (**7i**). Melting point: 207.6–208.9 °C. ^1H NMR (400 MHz, DMSO- d_6) δ 11.08 (s, 1H), 9.95 (s, 1H), 8.76 (s, 1H), 7.83 (s, 1H), 7.67 (d, $J = 8.5$ Hz, 1H), 7.51 (d, $J = 8.6$ Hz, 2H), 7.33 (d, $J = 2.3$ Hz, 1H), 7.24 (dd, $J = 8.8, 2.3$ Hz, 1H), 6.94 (d, $J = 8.8$ Hz, 2H), 5.73–5.53 (m, 1H), 5.07 (dd, $J = 12.9, 5.4$ Hz, 1H), 3.69–3.54 (m, 2H), 3.48–3.38 (m, 6H), 3.17–3.08 (m, 6H), 2.73 (s, 1H), 2.63–2.53 (m, 6H), 2.25–2.15 (m, 4H), 2.00 (dd, $J = 10.5, 5.0$ Hz, 2H), 1.92–1.71 (m, 8H), 1.61–1.55 (m, 2H). ^{13}C NMR (151 MHz, DMSO- d_6) δ : 172.80, 170.07, 167.55, 166.97, 159.34, 158.58, 156.54, 155.22, 133.84, 124.88, 121.70, 117.75, 115.73, 107.87, 107.80, 55.84, 54.92, 52.47, 48.77, 48.60, 46.95, 45.25, 32.56, 32.10, 30.97, 30.31, 29.59, 27.30, 26.54, 24.80, 22.16. HRMS (ESI): (m/z): [$M + H$] $^+$ calcd for $\text{C}_{46}\text{H}_{54}\text{N}_{12}\text{O}_6$, 870.4289; found 871.4370. HPLC purity: 99.27%, retention time = 1.461 min.

5-(2-(4-(4-((8-Cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)acetyl)piperazin-1-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (**7j**). Melting point: 234.5–236.8 °C. ^1H NMR (400 MHz, DMSO- d_6) δ 11.08 (s, 1H), 9.95 (s, 1H), 8.76 (s, 1H), 7.83 (s, 1H), 7.69 (d, $J = 8.5$ Hz, 1H), 7.50 (d, $J = 8.5$ Hz, 2H), 7.37 (d, $J = 2.2$ Hz, 1H), 7.26 (dd, $J = 8.7, 2.3$ Hz, 1H), 6.94 (d, $J = 8.7$ Hz, 2H), 5.77–5.49 (m, 1H), 5.07 (dd, $J = 13.0, 5.3$ Hz, 1H), 3.74 (d, $J = 5.6$ Hz, 2H), 3.62 (t, $J = 5.9$ Hz, 4H), 3.50 (t, 4H), 3.29 (s, 2H), 3.12 (t, 4H), 2.60 (t, 4H), 2.28–2.15 (m, 2H), 2.09–1.95 (m, 2H), 1.95–1.81 (m, 2H), 1.80–1.69 (m, 2H), 1.63–1.50 (m, 2H). ^{13}C NMR (151 MHz, DMSO- d_6) δ 172.80, 170.07, 167.55, 166.97, 159.34, 158.58, 156.54, 155.22, 133.84, 124.88, 121.70, 117.75, 115.73, 107.87, 107.80, 55.84, 54.92, 52.47, 48.77, 48.60, 46.95, 45.25, 32.56, 32.10, 30.97, 30.31, 29.59, 27.30, 26.54, 24.80, 22.16. HRMS (ESI): (m/z): [$M + H$] $^+$ calcd for $\text{C}_{40}\text{H}_{43}\text{N}_{11}\text{O}_6$, 773.3398; found 774.3481. HPLC purity: 99.63%, retention time = 1.381 min.

Molecular Simulation. Molecule Docking. The crystal structure of the CDK6 kinase domain (PDB ID: 5L2I, 2.75 Å resolution) was prepared with the Protein Preparation Wizard available in Maestro (Schrodinger, Inc., version 10.2). Water molecules were removed, and hydrogen atoms were added. Model energy was minimized with a root mean square deviation (RMSD) of 0.3 Å using the OPLS-2005 force field. The ligands were prepared using LigPrep at pH 7.0 $\pm$ 2.0 for Epik and docked into the ATP binding domain with the grid length set to 20 Å.

Protein Docking. All protein docking was carried out using Rosetta 3.1. Input CDK6/YY173 complex models were obtained by molecule docking and the CRBN/pomalidomide complex model downloaded from the PDB (PDB: 4CI3, 3.50 Å resolution). Relevant structure coordinates were combined into a single file and prepared for docking using the Rosetta “docking_prepack_protocol” program. Local perturbation docking was performed using Rosetta with the following command line options:

- partners A_B -dock_pert 3 8 -ex1 -ex2aro -nstruct 1000

To assess the landscape of possible binding modes for CDK6, the top 200 lowest-scoring docking decoys were selected and the shortest Euclidean distance between selected atoms on YY173 and pomalidomide was calculated in Python script. PROTAC conformers were generated using Cyndi and aligned using Shafts. In addition, Maestro was used for the optimization of complex structure. The visualization software PyMOL [The PyMOL Molecular Graphics System, Version

1.8.6.0 Schrödinger, LLC] was used to analyze the binding modes between degraders and protein.

Biological Evaluation. Cell Viability Assay. Jurkat cells were evenly dispersed in 96-well plates at a density of 8000 cells per well and incubated at 37 °C and 5% CO₂ for 8 h.³⁵ Different concentrations of compounds were added to each well, and the cells were further incubated for 72 h. Subsequently, 10 μL of CCK8 solution was added to each well and the plates were incubated in a dark environment at 37 °C and 5% CO₂ for 2 h. The absorbance values at 450 nm were then measured using a microplate reader, and the IC₅₀ values were calculated using GraphPad Prism software.

Western Bolt. Jurkat cells were evenly dispersed in a six-well plate and cultured at 37 °C with 5% CO₂. Subsequently, different concentrations of compounds were added, followed by the addition of lysis buffer (RIPA:PMSF = 99:1) (Thermo). The resulting mixture was placed on an ice bath for 25 min and then centrifuged at 13,000 rpm for 30 min at 4 °C. The total protein concentration was determined using the BCA kit (Beyotime).

Protein samples were separated using 10% SDS-PAGE, transferred to a PVDF membrane, and blocked for 2 h at room temperature. The membrane was then incubated with primary antibodies overnight at 4 °C, followed by incubation with secondary antibodies for 2 h at room temperature. The primary antibodies included CDK1 (Abcam No. ab133327), CDK2 (Abcam No. ab32147), CDK4 (Abcam No. ab108357), CDK5 (Abcam No. ab40773), CDK6 (Abcam No. ab124821), CDK7 (Abcam No. ab256787), CDK8 (Abcam No. ab224828), Caspase-3 (Cell Signaling Technology No. 9665S), cleaved Caspase-3 (Cell Signaling Technology No. 9664S), PARP (Cell Signaling Technology No. 9542S), cleaved PARP (Cell Signaling Technology No. 9541S), c-myc (Selleck No. A5011), and p-Rb (Cell Signaling Technology No. 8516S).

Cell Apoptosis Analysis. Cells were seeded in a six-well plate and treated with different concentrations of compounds. After 24 h, the cells were collected and apoptosis was assessed using Annexin V-FITC/PI staining. The outcome was measured through flow cytometry, and the data were analyzed by using FlowJo.

Cell Cycle Analysis. After 24 h of compound treatment, cells were collected and washed with precold PBS. The cells were then placed in a 4 °C refrigerator overnight. Cell cycle distribution was assessed using a Cell Cycle Kit (Beyotime). All samples were analyzed by flow cytometry, and the data were analyzed by using FlowJo.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.4c00965>.

7a–7j-induced concentration-dependent degradation of CDK4/6 protein in Jurkat cells; experimental section of chemistry; ¹H NMR spectra, ¹³C NMR spectra, and HPLC and HRMS analyses of representative compounds; and evaluation of the pharmacokinetic properties of 7f in rats (PDF)

PDB data on proteins (PDB)

List of compounds (CSV)

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Author Contributions

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Author Contributions

H.H., X.Z., J.W., and Q.L. are co-first authors and contributed equally to this work. H.H., Q.L., and L.Z. were in charge of compound synthesis. J.W. performed the computational work. X.Z. and L.C. performed the related biological experiments. Z.C., H.L., and Z.Z. conceived and supervised the overall project. The manuscript was written through contributions of all authors.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Finn, R. S.; Martin, M.; Rugo, H. S.; Jones, S.; Im, S. A.; Gelmon, K.; Harbeck, N.; Lipatov, O. N.; Walshe, J. M.; Moulder, S.; Gauthier, E.; Lu, D. R.; Randolph, S.; Diéras, V.; Slamon, D. J. Palbociclib and letrozole in advanced breast cancer. *N. Engl. J. Med.* **2016**, 375 (20), 1925–1936.

- (2) Hydbring, P.; Malumbres, M.; Sicinski, P. Non-canonical functions of cell cycle cyclins and cyclin-dependent kinases. *Nat. Rev. Mol. Cell Biol.* **2016**, *17* (5), 280–292.
- (3) Pernas, S.; Tolane, S. M.; Winer, E. P.; Goel, S. CDK4/6 inhibition in breast cancer: current practice and future directions. *Ther. Adv. Med. Oncol.* **2018**, *10*, 1758835918786451.
- (4) Cheng, W.; Yang, Z.; Wang, S.; Li, Y.; Wei, H.; Tian, X.; Kan, Q. Recent development of CDK inhibitors: An overview of CDK/inhibitor co-crystal structures. *Eur. J. Med. Chem.* **2019**, *164*, 615–639.
- (5) Ding, L.; Cao, J.; Lin, W.; Chen, H.; Xiong, X.; Ao, H.; Yu, M.; Lin, J.; Cui, Q. The roles of cyclin-dependent kinases in cell-cycle progression and therapeutic strategies in human breast cancer. *International journal of molecular sciences* **2020**, *21* (6), 1960.
- (6) Finn, R. S.; Alesin, A.; Slamon, D. J. Targeting the cyclin-dependent kinases (CDK) 4/6 in estrogen receptor-positive breast cancers. *Breast Cancer Res.* **2016**, *18*, 17.
- (7) de Melo Gagliato, D.; C Buzaid, A.; Perez-Garcia, J. M.; Llombart, A.; Cortes, J. Cdk4/6 inhibitors in hormone receptor-positive metastatic breast cancer: Current practice and knowledge. *Cancers* **2020**, *12* (9), 2480.
- (8) Xi, J.; Ma, C. X. Sequencing endocrine therapy for metastatic breast cancer: what do we do after disease progression on a CDK4/6 inhibitor? *Curr. Oncol. Rep.* **2020**, *22*, 57.
- (9) Groenland, S. L.; Martínez-Chávez, A.; van Dongen, M. G.; Beijnen, J. H.; Schinkel, A. H.; Huitema, A. D.; Steeghs, N. Clinical pharmacokinetics and pharmacodynamics of the cyclin-dependent kinase 4 and 6 inhibitors palbociclib, ribociclib, and abemaciclib. *Clinical pharmacokinetics* **2020**, *59* (12), 1501–1520.
- (10) Powell, K.; Prasad, V. Concerning FDA approval of trilaciclib (Cosela) in extensive-stage small-cell lung cancer. *Translational oncology* **2021**, *14* (11), No. 101206.
- (11) Dhillon, S. Trilaciclib: first approval. *Drugs* **2021**, *81* (7), 867–874.
- (12) Wang, Y.; Chen, X.; Yan, Y.; Zhu, X.; Liu, M.; Liu, X. Discovery and SARs of 5-Chloro-N 4-phenyl-N 2-(pyridin-2-yl) pyrimidine-2, 4-diamine Derivatives as Oral Available and Dual CDK 6 and 9 Inhibitors with Potent Antitumor Activity. *J. Med. Chem.* **2020**, *63* (6), 3327–3347.
- (13) Knudsen, E. S.; Witkiewicz, A. K. The strange case of CDK4/6 inhibitors: mechanisms, resistance, and combination strategies. *Trends in cancer* **2017**, *3* (1), 39–55.
- (14) Niu, Y.; Xu, J.; Sun, T. Cyclin-dependent kinases 4/6 inhibitors in breast cancer: current status, resistance, and combination strategies. *Journal of Cancer* **2019**, *10* (22), 5504.
- (15) Pandey, K.; An, H. J.; Kim, S. K.; Lee, S. A.; Kim, S.; Lim, S. M.; Kim, G. M.; Sohn, J.; Moon, Y. W. Molecular mechanisms of resistance to CDK4/6 inhibitors in breast cancer: A review. *International journal of cancer* **2019**, *145* (5), 1179–1188.
- (16) Pancholi, S.; Ribas, R.; Simigdala, N.; Schuster, E.; Nikitorowicz-Buniak, J.; Ressa, A.; Gao, Q.; Leal, M. F.; Bhamra, A.; Thornhill, A.; Morisset, L.; Montaudon, E.; Sourd, L.; Fitzpatrick, M.; Altelaar, M.; Johnston, S. R.; Marangoni, E.; Dowsett, M.; Martin, L.-A. Tumour kinome re-wiring governs resistance to palbociclib in oestrogen receptor positive breast cancers, highlighting new therapeutic modalities. *Oncogene* **2020**, *39* (25), 4781–4797.
- (17) Pandey, K.; Lee, E.; Park, N.; Hur, J.; Cho, Y. B.; Katuwal, N. B.; Kim, S. K.; Lee, S. A.; Kim, I.; An, H. J.; Hwang, S.; Moon, Y. W. Deregulated immune pathway associated with palbociclib resistance in preclinical breast cancer models: Integrative genomics and transcriptomics. *Genes* **2021**, *12* (2), 159.
- (18) Sun, X.; Wang, J.; Yao, X.; Zheng, W.; Mao, Y.; Lan, T.; Wang, L.; Sun, Y.; Zhang, X.; Zhao, Q.; Zhao, J.; Xiao, R.-P.; Zhang, X.; Ji, G.; Rao, Y. A chemical approach for global protein knockdown from mice to non-human primates. *Cell Discovery* **2019**, *5* (1), 10.
- (19) Zhou, F.; Chen, L.; Cao, C.; Yu, J.; Luo, X.; Zhou, P.; Zhao, L.; Du, W.; Cheng, J.; Xie, Y.; Chen, Y. Development of selective mono or dual Protac degrader probe of CDK isoforms. *Eur. J. Med. Chem.* **2020**, *187*, No. 111952.
- (20) Bedford, L.; Lowe, J.; Dick, L. R.; Mayer, R. J.; Brownell, J. E. ubiquitin-like protein conjugation and the ubiquitin–proteasome system as drug targets. *Nat. Rev. Drug Discovery* **2011**, *10* (1), 29–46.
- (21) Adams, J. The proteasome: a suitable antineoplastic target. *Nature Reviews Cancer* **2004**, *4* (5), 349–360.
- (22) Khan, S.; He, Y.; Zhang, X.; Yuan, Y.; Pu, S.; Kong, Q.; Zheng, G.; Zhou, D. Proteolysis TArgeting Chimeras (PROTACs) as emerging anticancer therapeutics. *Oncogene* **2020**, *39* (26), 4909–4924.
- (23) Zhao, B.; Burgess, K. PROTACs suppression of CDK4/6, crucial kinases for cell cycle regulation in cancer. *Chem. Commun.* **2019**, *55* (18), 2704–2707.
- (24) Wang, C.; Zhang, Y.; Wu, Y.; Xing, D. Developments of CRBN-based PROTACs as potential therapeutic agents. *Eur. J. Med. Chem.* **2021**, *225*, No. 113749.
- (25) Anderson, N. A.; Cryan, J.; Ahmed, A.; Dai, H.; McGonagle, G. A.; Rozier, C.; Benowitz, A. B. Selective CDK6 degradation mediated by cereblon, VHL, and novel IAP-recruiting PROTACs. *Bioorg. Med. Chem. Lett.* **2020**, *30* (9), No. 127106.
- (26) Rana, S.; Bendjennat, M.; Kour, S.; King, H. M.; Kizhake, S.; Zahid, M.; Natarajan, A. Selective degradation of CDK6 by a palbociclib based Protac. *Bioorganic & medicinal chemistry letters* **2019**, *29* (11), 1375–1379.
- (27) Steinebach, C.; Ng, Y. L. D.; Sosić, I.; Lee, C. S.; Chen, S.; Lindner, S.; Vu, L. P.; Bricelj, A.; Haschemi, R.; Monschke, M.; Steinwarz, E.; Wagner, K. G.; Bendas, G.; Luo, J.; Gütschow, M.; Krönke, J. Systematic exploration of different E3 ubiquitin ligases: an approach towards potent and selective CDK6 degraders. *Chem. Sci.* **2020**, *11* (13), 3474–3486.
- (28) Su, S.; Yang, Z.; Gao, H.; Yang, H.; Zhu, S.; An, Z.; Wang, J.; Li, Q.; Chandarlapaty, S.; Deng, H.; Wu, W.; Rao, Y. Potent and preferential degradation of CDK6 via proteolysis targeting chimera degraders. *J. Med. Chem.* **2019**, *62* (16), 7575–7582.
- (29) Jiang, B.; Wang, E. S.; Donovan, K. A.; Liang, Y.; Fischer, E. S.; Zhang, T.; Gray, N. S. Development of dual and selective degraders of cyclin-dependent kinases 4 and 6. *Angew. Chem., Int. Ed.* **2019**, *58* (19), 6321–6326.
- (30) Gu, S.; Cui, D.; Chen, X.; Xiong, X.; Zhao, Y. PROTACs: an emerging targeting technique for protein degradation in drug discovery. *BioEssays* **2018**, *40* (4), 1700247.
- (31) Nowak, R. P.; DeAngelo, S. L.; Buckley, D.; He, Z.; Donovan, K. A.; An, J.; Safaei, N.; Jedrychowski, M. P.; Ponthier, C. M.; Ishoe, M.; Zhang, T.; Mancias, J. D.; Gray, N. S.; Bradner, J. E.; Fischer, E. S. Plasticity in binding confers selectivity in ligand-induced protein degradation. *Nat. Chem. Biol.* **2018**, *14* (7), 706–714.
- (32) Fassl, A.; Geng, Y.; Sicinski, P. CDK4 and CDK6 kinases: From basic science to cancer therapy. *Science* **2022**, *375* (6577), eabc1495.
- (33) Liu, X.; Bai, F.; Ouyang, S.; Wang, X.; Li, H.; Jiang, H. Cyndi: a multi-objective evolution algorithm based method for bioactive molecular conformational generation. *BMC Bioinf.* **2009**, *10*, 101.
- (34) Liu, X.; Jiang, H.; Li, H. SHAFTS: a hybrid approach for 3D molecular similarity calculation. 1. Method and assessment of virtual screening. *J. Chem. Inf. Model.* **2011**, *51* (9), 2372–2385.
- (35) Yang, Y.; Shi, X.; Chen, Z.; Xu, Y.; Qian, X.; Zhu, W. Novel seven-membered ring-fused naphthalimide derivatives with potentials for cancer theranostics. *Chin. Chem. Lett.* **2023**, *34* (4), 107696.